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Asset Information Management

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Asset Information Management

Asset Management, Glencore Copper
30 May 2022, Version 1.0

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Preface

The Asset Management Framework, Policy and Manuals are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities that advance everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this manual will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

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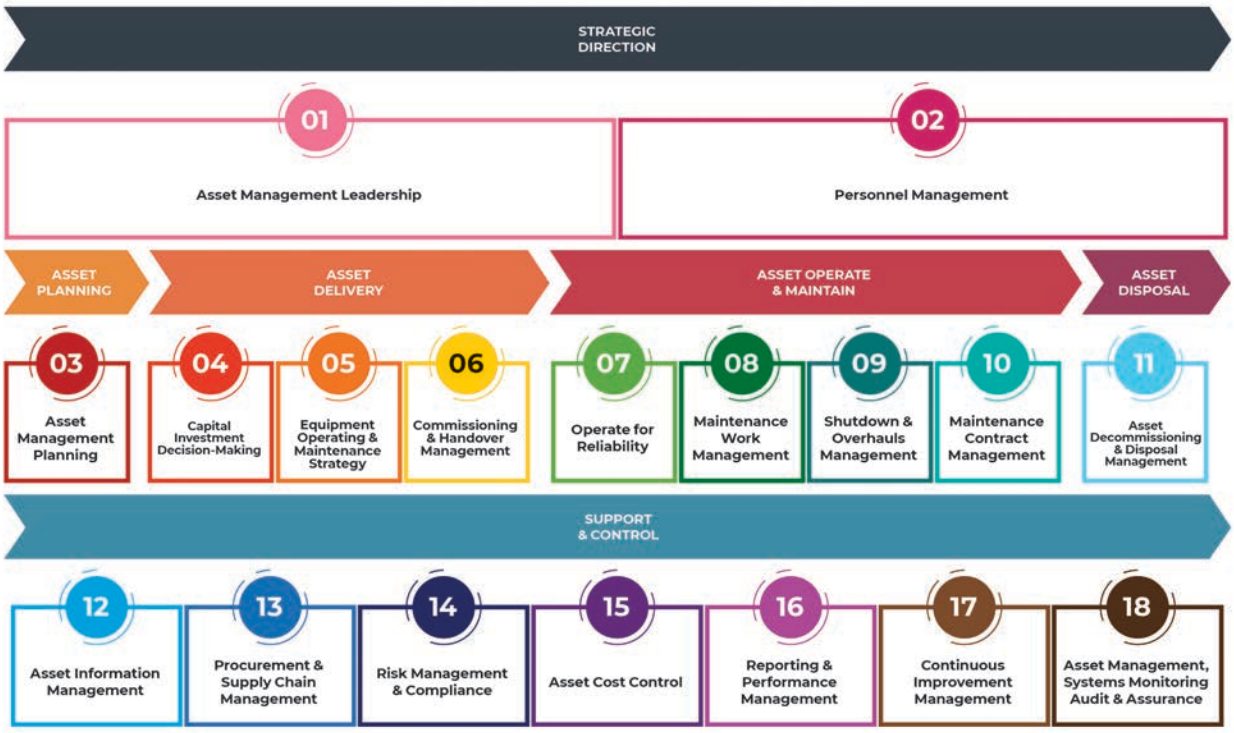
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1. Introduction

The Asset Management Framework, Policy and Manuals define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring assets are capable of meeting business and stakeholder objectives.

Figure 1-1 describes the high-level Asset Management Framework, which enables Glencore Copper to deliver its Asset Management purpose and vision.

FIGURE 1-1 ASSET MANAGEMENT FRAMEWORK



2. Objectives

The primary objectives of the Asset Information Management Manual are to ensure:

- Long-term optimised approach to management of asset information derived from and consistent with, the organisational strategic plan and other document and data management standards.
- Mandatory legislation requirements for asset information are being met.
- Asset information is accurate, complete, valid, consistent, uniquely named, accessible, and secure.
- Ongoing monitoring, improvement, and governance of asset information to ensure changes to quality and issues are identified and addressed.

2.1 Accountabilities

Dedicated resources must be assigned, and capabilities developed within each Operation to deliver Asset Information Management processes that enable effective delivery of the Asset Management objectives.

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Assign single point accountability for each or all of the following:

- Asset Information Standards
- Asset Information Configuration and Support
- Computerised Maintenance Management Systems (CMMS) Establishment and governance
- Asset Management Data Accuracy and Completeness

2.2 Scope

Asset information is a combination of data about physical assets and how it is used to inform decisions about how physical assets are managed both for short term operational purposes and for long term strategic planning. Asset information covers:

- Asset Register/Hierarchy
- Asset History
- Asset Documents
- Design Information (as built and modification)
- Images and Multimedia

Appropriate quality data and information are key to the management of Glencore Copper's Asset Management Framework. It provides support for the asset management activity itself, for policy making and for supporting both operational and strategic business decisions.
Glencore Copper shall establish and maintain systems that manage asset information. The systems shall be designed to provide sufficient support and information to meet the Glencore Copper's asset management decision making objectives and integrated interfaced with CMMS where practical and justifiable,
The scope of asset information is significant and is directly related to the scope of the Asset Management Framework and the effective utilisation of the CMMS such as SAP calculations for Asset Management metrics.

3. Definitions

Asset Register – A list or set of related and/or hierarchical codes and associated descriptions which ultimately define and describe the individual items and/or components of equipment installed in the plant.

Assurance - the processes that assess the level of compliance with standards and the effectiveness and efficiency of business policies and activities.

Compliance- the term for the task of documenting that activities and outputs conform to set legislation, standards and company procedures and policies.

Data - numbers, words, symbols, pictures, etc. without context or meaning, i.e., data in a raw format, e.g. 25 metres.

Information - a collection of data expressed with a supporting context e.g. The span of the bridge is 25 metres.

Information Management - the means by which an organisation maximises the efficiency with which it plans, collects, organises, uses, controls, stores, disseminates, and disposes of its Information, and through which it ensures the value of information is identified and exploited to the maximum extent possible. The aim has often been described as getting the right information to the right person, in the right format and medium, at the right time.

Information Technology - the technology used (e.g. applications and software) to support business functions and processes.

Knowledge - a combination of experience, values, information in context, and insight that form a basis for decision making. It refers to the process of comprehending, comparing, judging, remembering, and reasoning.

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Master Data – data that is static in nature (seldom changed) and is reference data shared or exchanged across strategic applications, work processes, lines of business, or enterprises.

Record – evidence in the form of information representing an account of something that has occurred e.g. a maintenance record detailing an item of work being carried out.

Requirements - the organisational need that shall be fulfilled.

Deadline Monitoring - is a workflow runtime system function that monitors the start and end deadlines for the processing of selected work items in an approval process.

SAP Maintenance Strategy - defines the rules for the sequence of planned maintenance work. It contains general scheduling information and can therefore be assigned to as many maintenance task lists (PM task lists) and maintenance plans as required. A maintenance strategy contains maintenance packages in which the following information is defined:

- The cycle in which the individual work should be performed (for example, every two months, every 5,000 km, every 500 operating hours)
- Other data which affects scheduling.

Standards - Statements of how data should be captured, work activities undertaken etc. Standards can be internal, organisation specific documents or external, regulatory/legal requirements to be complied with.

Specifications - Statements of how data shall be gathered, processed, and recorded.

Transactional Data - data that is dynamic in nature and refers to the transaction that are carried out on a Master Data object, i.e., work order on an item of Equipment.

4. Approach

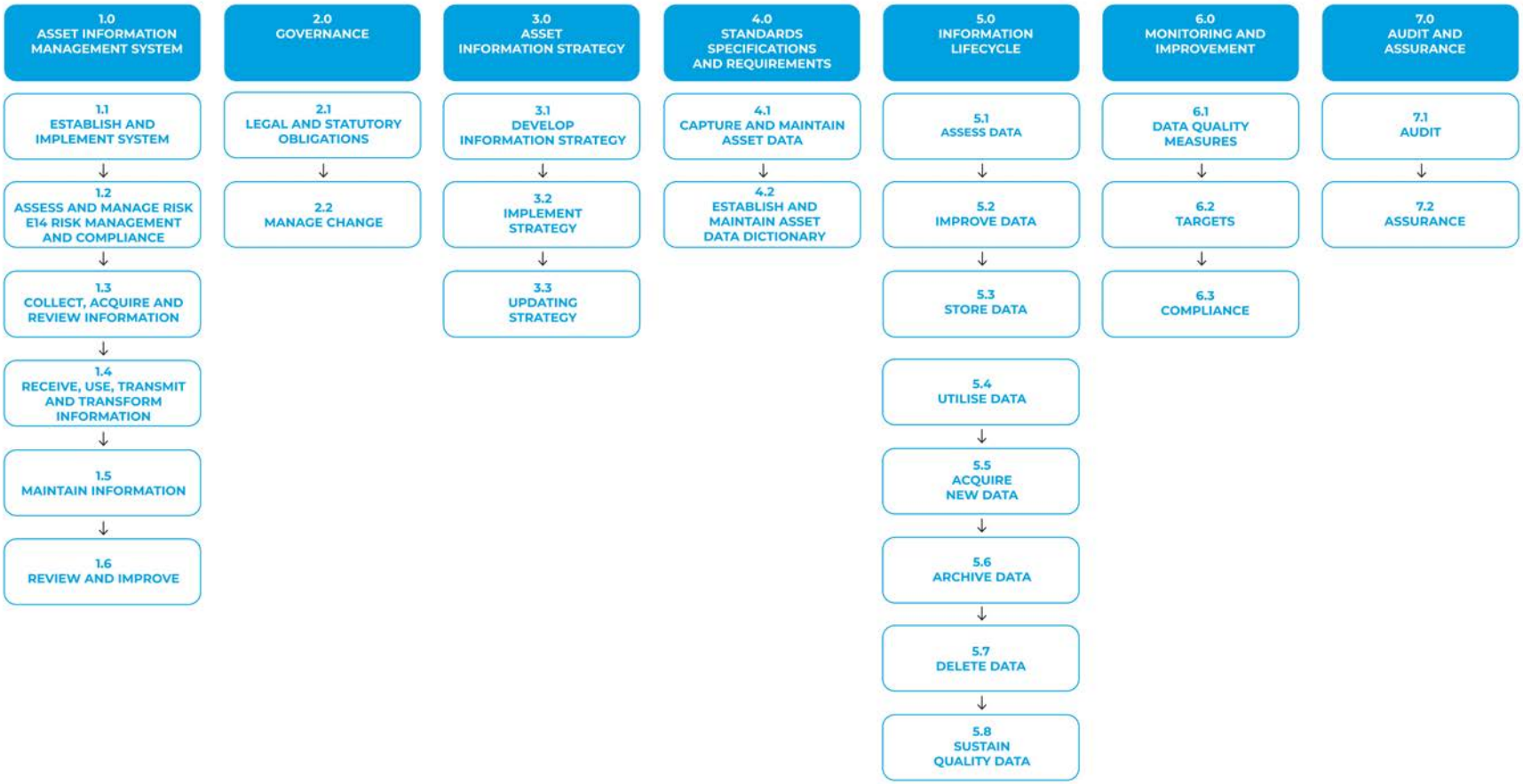
The approach for Asset Information Management is described through:

- Process Flow and Descriptions - pictorially illustrates the key Asset Information Management activities.
- Process RACI Matrix - indicates for each activity, the Functions who are Responsible, the Function that is Accountable (should be only one), the Functions who are Consulted prior to final decision/action, and those who are Informed of any outcomes and actions, refer to Section 4.3

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4.1 Process Flow

FIGURE 4-1 ASSET INFORMATION MANAGEMENT PROCESS



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4.2 Process Description

TABLE 4-1 ASSET INFORMATION MANAGEMENT PROCESS DESCRIPTION

PROCESS STEP	DESCRIPTION
1.0 Asset Information Management System	**Is to define and manage the use of asset information by Glencore Copper sites as a key component, and in support of, the Asset Management Framework.**
1.1 Establish and Implement CMMS	Establishing and implementing CMMS to manage assets: • Implementing the CMMS to a documented standard that meets requirements. • Programs are version-controlled and access to modify plant settings restricted. • Establish a CMMS Data Structure • Asset Register and CMMS configuration are aligned, in so far as, the configuration of the CMMS is done so in a manner which enables business objectives to be carried out, as opposed to accepting an off the shelf CMMS solution.
1.2 Assess and Manage Risk	Glencore Copper sites shall assess the business risks of not having particular asset information available at the right quality to the business and identify the criticality of different information sets. Glencore Copper sites shall implement appropriate mitigation and control measures to ensure resilience and business continuity in relation to critical information. Risk shall be managed in consultation with E14 Risk Management and Compliance.
1.3 Collect, Acquire and Review Information	Glencore Copper sites shall as part of the Glencore Copper Asset Management Framework (AMF) improvement, periodically collect, acquire and review new information to meet changing business needs, this could include: • Bulk transfer of legacy data into new systems • Register of assets not previously registered • Collect data attributes required by the business • Receive and store information which has been identified in project planning phase as projects are handed over. Information is archived as per process step 5.6 in this manual. • Establishing systems to communicate equipment and process status, trends, and alarms, providing adequate warning and effectiveness reviews. • Establishing systems to capture delays, losses, and downtime reporting. Glencore Copper Time Usage Model (TUM) as an example. • Modifications to asset systems, plant, and equipment.
1.4 Receive, Use, Transmit and Transform Information	Glencore Copper sites shall ensure sufficient access is granted and made available to asset information, so users are fully supported to enable them to carry out their duties. Each site shall define and implement accessibility to asset information. Each site shall determine requirements for individuals within functional groups. Glencore Copper sites shall receive and record relevant information as part of the ongoing development, implementation and improvement of the AMF and business activities. When asset information is transformed or converted into other information forms or formats, that the process and output ensure the transformation is consistent with and supports the Asset Management Framework.
1.5 Maintain Information	• Data Quality KPI's shall be established. • Sufficient care is applied by users in maintaining the quality of data in line with the data quality KPI requirements. • Accessibility and information security policy and procedures are defined and implemented to reflect business requirements.

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	• The security environment is supported by management and understood and enforced across the user community, including contractors. • The exchange of critical information being restricted to ensure confidential information of the organization is not passed to unauthorised persons (but without withholding access to it from business users who have a genuine need to use it). • Defining access and ownership of information generated by contractors, suppliers and Original Equipment Manufacturers (OEMs). • Structuring and access for storing and updating of server-based engineering information.
1.6 Review and Improve	In conjunction with new requirements that have been described in the Asset Management Plans, Glencore Copper sites shall identify opportunities for improving the Asset Information Management System in support of the overall Asset Management Framework.
2.0 Governance	**Asset information governance ensures control and oversight of the standards, policies, processes, roles, and responsibilities that direct overall management of Glencore Copper sites asset information. Governance ensures asset information is accurate, consistent, complete, available, and secure.**
2.1 Legal and Statutory Obligations	Glencore Copper sites shall ensure the asset information management system is compliant with applicable legal requirements. Consideration should be given to current and developing corporate policy, regional, national, and international legislation. All documents shall be identified, and formats established that meet statutory, legal and management requirements, and where required archived as per process step 5.6 of this manual.
2.2 Manage Change	Change shall be managed in line with the Management of Change approach in the Group Enterprise Risk Management Standard. As a minimum this will include: • Consultation with stakeholders and major end users about the need and benefits of change and how this might impact them. • Assess readiness for change and implement change at the appropriate time. • Identify the impact of change and individuals impacted. • Train and support the individuals over a period of time until the change is embedded. • Ensure staff who are responsible for change are competent to do so. • Ensure all upstream and downstream aspects of the change are considered and implemented. • Ensure all asset and/or engineering information that may be linked to the change, is closed out as part of the Management of Change process Follow site authority levels and Management of Change processes and procedures.
3.0 Asset Information Strategy	**Data is an asset which requires a similar strategic approach and management activities the overarching Asset Management Framework provides.**
3.1 Develop Information Strategy	Glencore Copper sites shall develop an Information Strategy reference document to inform the organisation regarding: • People – skills and capabilities required to develop and deliver the information strategy. • Technology – systems and tools required to support the implementation of the strategy. • Process – mechanisms for managing data and creating value from it. • Data – what data is critical and what are the regulatory and legal obligations.
3.2 Implement the Information Strategy	Measure current Information Strategy performance to understand the current governance mechanisms, the current state of data quality, the existing standards, and policies. This will enable the understanding of the gap to deliver on the strategy.
3.3 Update Strategy	The Asset Information Strategy shall be reviewed annually and updated periodically and site management of changed process and procedures followed.
4.0 Standards, Specifications and Requirements	**It is important for Glencore Copper sites to have a clear and agreed view of the asset information required both for asset management purposes and where asset information supports other business processes.** Information is needed to support business processes such as: **Strategic** – at a level to support a "Strategic Objectives", for example measures of corporate business drivers of safety, performance, environment, etc.

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	Tactical – to support "Technical Policy Development" (e.g., data and information to support analysis of maintenance intervention policies) **Operational** – to deliver asset related work through the efficient scheduling of resources.
4.1 Capture and Maintain Asset Data	Glencore Copper have various systems for capturing and maintaining data such as: • Asset hierarchy/list • Operating and maintenance manuals • Technical drawings which may include: ○ Process Flow Diagrams (PFD's) ○ Process and Instrumentation Diagrams (P&ID's) ○ General Arrangements ○ Detail Drawings, i.e. Civil, Structural, Mechanical, Piping, Electrical and Control ○ Spare Parts and components • Technical specifications • Functional descriptions • Asset management plans • Critical component list • Work management data (refer to AMF E08 Maintenance Work Management) • Performance data (refer to AMF E16 Reporting and Performance Management) • Condition and residual life data • Acquisition, operation, and maintenance cost data • Online technical information for example OEM manuals, parts list etc. • Spare parts list and interchangeability records • Manufacturer Data Reports (MDR) Consideration should be made for: • **Frequency of use** – how often the information is needed in order to support the business process and decision making • **Type** – the type of data required for a business purpose • **Specific attributes and units** – e.g. the age of an asset in years • **Precision and accuracy** – e.g. how exact measurements need to be and to what extent can inaccuracy be tolerated.
4.2 Establish and Maintain Asset Data Dictionary	Establish and maintain an Asset Data Dictionary with all relevant asset standards, specifications, and requirements. This specifies a set of common characteristics for Asset data, which must be adhered to for the data to be considered fit for a specific purpose. An Asset Data Dictionary specifies: • How assets relate to each other, which is in the form of a hierarchy. Different levels in the hierarchy will be defined, similarly, the level of detail at which the asset hierarchy stops are specified. • How assets are classified, i.e. grouped by generic type. • How assets are defined by function, i.e. generic purpose or duty • Where assets are located • The attributes relevant to a particular class of asset and the definition of these attributes; and • Rules for how to define assets in different circumstances. The Glencore Copper SAP Master Data Design Standards shall be referred to for the purpose of the Dictionary development.
5.0 Information Life Cycle	**Information of any kind has its own life cycle. Data is created, used, maintained and archived when no longer pertinent. For asset information, this lifecycle should mirror the lifecycle of the physical assets the information describes.**
5.1 Assess Data	Assess the current state of asset information and data including but not limited to: • Accuracy – does the data correctly represent the asset it relates to? • Validity – is the data stored in the correct format? • Completeness – Are all assets and required attributes populated? For SAP Master Data refer to Section 5 of this manual which provides information regarding master data object and their relationships. • Consistency – Does the same asset have the same identifier across data sets?

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	• Uniqueness – Is each asset recorded once and once only? • Timeliness – What is the time delay between a change to an asset and the corresponding data change? Timelines for information and data changes are to comply to site requirements. Any assessment of data quality should be made against the data requirements in the Data Dictionary. Data is an asset and without due care and attention will degrade. Degradation can take place for many reasons including: • Not being kept up to date • Not being accurate or complete when it is updated • Information on assets being referenced differently in systems • Error when entering data • Use of different terminology or descriptions • Poor data migration from legacy systems • Business or technology changes rendering the data obsolete • Poor governance leading to lack of control regarding system user access and lack of training. • Inconsistent or non-structured application of data and information and systems • Poor application of Management of Change process This could lead to negative outcomes for Glencore Copper sites such as: • Asset failure • Increased exposure to risk • Poor asset performance • Increase in cost • Operational losses • Noncompliance to legal and statutory requirement leading to financial loss or possible licence to operate.
5.2 Improve Data	Periodically assess whether information is adequate to meet business requirements for effective decision making, and whether the results are being achieved as a result of decisions using this information are delivering the expected outcomes. This will provide: • Consistently up-to-date asset management data including asset numbering, technical data, catalogued spares, Bills of Materials, standard maintenance and operating tasks, safety files, drawings, change management, costs etc. • Information provided in reports is based on a consistent data structure, enabling better identification of improvement opportunities. • Purchase of plant and equipment is conducted using the latest standards. • Critical and sensitive information is secure and up to date. • An auditable trail of asset data changes.
5.3 Store Data	The CMMS is the primary location for the accessibility of asset data and information. This is either stored directly or linked via other Information Systems (IS). The asset information and data contained within the CMMS provides the foundational data that is necessary for: • Asset management decision making • Maintenance Work Management, refer Element E08 • Cost Control, refer Element E15 • Equipment Operating and Maintenance Strategy, refer Element E05 • Project Management • Procurement and Supply Chain Management, refer Element E13 • Maintenance Contracting and Outsourcing, refer Element E10 • Shutdown and Overhaul Management, refer Element E09 Engineering documentation may be stored in other IS but where appropriate it should be linked to the Asset in the CMMS. Care should be taken to ensure there is only one source of the truth.

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5.4 Utilise Data	Data shall be of sufficient quality and comply with current policies, processes, standards and procedures and it can be used for risk-based decision making. Data collected shall provide value in its use and not be collected as a capture all approach. Data is primarily used for: • Asset Maintenance – ensuring the right maintenance is done on the right asset at the right time at the right cost in line with financial forecasts. • Asset Replacement – ensuring assets are replaced at end of useful life. • Regulatory Reporting – ensuring the organisation is meeting its statutory and regulatory responsibilities. • Inventory and Valuation – ensuring the organisation has the correct inventory at hand and understands its value. • Asset Risk – ensuring risk is managed in an appropriate manner and data is of the right quality to enable risk-based decisions. Risks include, safety, environment, financial and asset performance.
5.5 Acquire New Data	Acquiring Asset Data can come from various sources including: • Asset Acquisition - Asset information is acquired during asset design, installation and commissioning and repair/replacement. • Surveys – physical surveys of assets provide a clear information stream to populate data models, for example asset condition to estimate residual life for Asset Management Planning purposes. • Transfers – Asset information may be transferred from other business units or within the same business unit. All acquisition processes require a clear understanding of data specifications, the process for acquiring data and where the data will be stored. Issues of accuracy and reliability are relevant for each data item, with coverage and population being relevant for the asset inventory as a whole. The Capital Management Guidelines (Major, Intermediate, Minor) shall be consulted in the acquisition process. Acquiring asset data shall be conducted in parallel to construction/commissioning activities. Key considerations are: • Information provisions are as important as the assets themselves. • Data is acquired from design and procurement information. • Easy access to the Data Dictionary. • No project is to be handed over until the information identified in the project planning stage is handed over to the asset owner. • Clear focus on information governance.
5.6 Archive Data	A Data Archive strategy shall be developed, items to consider include: • What should be archived and why? • For how long should records be retained? • What are the obligations for retaining data? • Risk associated with not been able to retrieve information? • How quickly does information need to be retrieved? • How to access archived information? • Security regarding archived information? • How data is to be archived? • What is the value of retaining data and how this changes over time?
5.7 Delete Data	How long to keep information for shall form part of the Archiving Strategy and records to be deleted in accordance with that strategy.
5.8 Sustain Quality Data	Sustaining good data involves: • Understanding what the clear data requirements are • Monitoring of the data processes and the data itself • Improvement to behaviours and governance • Audit to determine what is actually happening
6.0 Monitoring and Improvement	**The monitoring process enables the key areas of information and key data to be monitored. It provides visibility to weaknesses in the data and software which would**

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	otherwise be hidden and unaddressed until the risk they represent is realised when used in practise.
6.1 Data Quality Measures	Data shall conform to the following criteria: • Accuracy – the data fully represents the asset it relates to. • Completeness – all assets and attributes are populated. • Validity – data is stored in the correct format. • Consistency – assets have the same identifier across all systems, standards, and processes. • Timeliness – minimal time to update asset information. • Uniqueness – no replication of data and information. There shall be regular reviews of data and information to ensure it still represents the current state and should also be assessed to ensure systems, policies, process, and procedures can accommodate for the future state. A Data Security Policy shall be developed and implemented as per Glencore Copper information Governance Policy, Information Security Policy, Data Protection Standard and End-User IT Services Agreement.
6.2 Targets	Relevant targets shall be established to quickly identify whether data improvement performance is acceptable. This could be represented by a simple traffic light process regarding the different data types. For example, SAP Plant Maintenance Mater Data what is the level of data quality as defined by process step 6.1 of this manual.
6.3 Compliance	Glencore Copper sites shall comply to the following standards: • Legal for recording data and information • Regulatory for information • Interoperability standards to allow systems and organisations to share information • Technical standards • Internal policy and quality standards
7.0 Audit and Assurance	**Audit and assurance enable Glencore Copper sites to review our data, software, and processes objectively against standards and requirements. Using this approach, we can evaluate if our data fits within expected ranges and meets its intent. As an organisation with a mature and effective asset information management system will gain greater confidence in our data.**
7.1 Audit	Data management activities will deliver good and improving data, and the ongoing governance and monitoring of data will ensure changes to quality and new issues are identified and addressed. Audits are to be conducted routinely. There are many reference sources and standards for auditing activities, but the following key factors shall be considered: • Sample size to include enough assets and attributes to ensure there is statistical significance in the results; if discrepancies are identified, an even larger sample may be viewed to determine the magnitude of the discrepancies. • The selection of assets to audit should be randomised (to ensure geographic coverage is unbiased) and stratified (to ensure the proportion of, say, asset classes audited is comparable to the proportion of total assets). The randomised selection is to cover assets across all criticality types A, B, and C. As a guiding principle the sample data size should cover 50% of A, 30% of B and 20% of C critical items. Further analysis may be required subject to discovered data issues. • Auditors should be familiar with the technology of the assets being audited. • Data quality problems are often the symptoms of other business-related issues. Auditors should attempt to determine the true root cause(s) of data quality issues.

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	Areas to audit include: • **Data audit** – Formal audit of the quality of data to current asset information standards. • **Process audit** – Audit of business processes that use/ create/ update / interact with asset data. Audits should also ensure process roles and responsibilities are understood and carried out correctly and all relevant standards and legislation are complied with. • **Culture audit** – Audit of organisational culture to assess whether a supportive culture exists, to identify cultural barriers and competing pressures which adversely affect asset data quality.
7.2 Assurance	Assurance activities to include audit, ongoing measurement of data accuracy and establishing reports to measure the effectiveness of data update processes. The outputs of assurance activities should be one of the inputs into overall governance processes.

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4.3 Process RACI Description

The RACI matrix below in Table 4-2 addresses the high-level functional elements of the Asset Information Management process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

Note: The HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-2 PROCESS RACI

R	Responsible	The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.
A	Accountable	The "owner" –the individual who ultimately "owns" the activity – there should only be one person.
C	Consulted	The individual(s) who needs to provide feedback and contribute to the activity.
I	Inform	The individual(s) who needs to be informed after a decision or action is taken if relevant. (via 1-way communication).

NOTES:

- Each functional group are only responsible for Asset Information under their control and will only be consulted or informed when the information relates to their function/role.
- The cross functional management functional group consist of Operations, Maintenance and Supply Chain Management. The maintenance function can also be implied to be the Asset Management function or similar if existing at site.
- The Asset Information Management Function generally covers a number of individuals who have the authority and competence to make formal changes to asset information and data such as CMMS Master data.

Process Step \ Function	Cross Functional Management	Reliability Management	Maintenance Execution	Supply / Warehouse	Maintenance Planning	Technical Services	Asset Information Management
1.0 Asset Information Management System							
1.1 Establish and Implement CMMS	A	C	R	C	R	C	C
1.2 Assess and Manage Risk	A	R	C	R	C	R	C
1.3 Collect and Acquire Information	A	R	R	R	R	R	R
1.4 Receive, Use, Transmit and Transform Information	A	R	R	R	R	R	R
1.5 Maintain Information	A	R	R	R	R	R	R
1.6 Review and Improve	A	R	R	R	R	R	R
2.0 Governance							
2.1 Legal and Statutory Obligations	A	R	C	R	C	R	R
2.2 Manage Change	A	R	C	R	C	R	R
3.0 Asset Information Strategy							

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Function / Process Step	Cross Functional Management	Reliability Management	Maintenance Execution	Supply / Warehouse	Maintenance Planning	Technical Services	Asset Information Management
3.1 Develop Information Strategy	A	R	I	R	I	R	C
3.2 Implement Strategy	A	R	I	R	I	R	R
3.3 Update Strategy	A	R	I	R	I	R	R
4.0 Standards, Specifications and Requirements							
4.1 Capture and Maintain Asset Data	A	R	R	R	R	R	R
4.2 Asset Data Dictionary	A	R	R	I	R	R	R
5.0 Information Lifecycle							
5.1 Assess Data	A	R	R	R	R	R	R
5.2 Improve Data	A	R	R	R	R	R	R
5.3 Store Data	A	C	C	C	C	C	R
5.4 Utilise Data	A	R	R	R	R	R	R
5.5 Acquire New Data	A	C	C	C	C	C	R
5.6 Archive Data	A	C	C	C	C	C	R
5.7 Delete Data	A	C	C	C	C	C	R
5.8 Sustain Quality Data	A	C	C	C	C	C	R
6.0 Monitoring and Improvement							
6.1 Data Quality Measures	A	R	R	R	R	R	R
6.2 Targets	A	R	R	R	R	R	R
6.3 Compliance	A	R	R	R	R	R	R
7.0 Audit and Assurance							
7.1 Audit	A	R	C	R	C	R	C
7.2 Assurance	A	R	C	R	C	R	C

TABLE 4-3 FUNCTIONAL DESCRIPTIONS

Function	Description
Cross Functional Management (Operations, Maintenance and Supply Chain)	• Ensures the establishment and implementation of the CMMS. Each site is to decide which function within the cross functional group is to take overall accountability or if a joint accountability. • Ensures risks are assessed and managed. • Ensures information is collected, acquired and reviewed as business needs change. • Ensures access is granted to asset information so users are fully supported to enable them to carry out their duties, this includes the receival, use, transmission and transformation of asset information. • Ensures asset information is maintained to ensure data quality, accessibility, security, confidentiality, ownership and is stored correctly and securely. • Ensures improvement opportunities are identified for the asset management information system. • Ensures control and oversight of standards, policies, processes, roles and responsibilities direct overall management of site asset information, ensuring legal

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	and statutory obligations are met and the Glencore Copper Management of Change process is followed. • Ensures an Asset Information Strategy is developed, implemented and updated periodically. • Ensures asset information supports strategical, tactical and operational business processes, asset data is captured and maintained, and an asset dictionary is established and maintained. • Ensures the fully life cycle of information and data is established, to enable data to be, assessed, improved, stored, utilised, acquired, archived, deleted and sustained. • Ensures data and information is monitored and improved, to ensure data is accurate, complete, valid, consistent, unique and updated in a timely manner. • Ensures targets are established to measure data improvement performance is acceptable and data is complaint to legal, regulatory, standards and policies. • Ensures asset information, data, process and systems are audited, and assurance is guaranteed.
Reliability Management	• Provides input into the establishment and implementation of the CMMS. • Responsible for managing and assessing risk. • Responsible for the provision that information is collected, acquired and reviewed as business needs change. • Responsible for the provision that access is granted to asset information, so users are fully supported to enable them to carry out their duties, this includes the receival, use, transmission and transformation of asset information. • Responsible for the provision that asset information is maintained to ensure data quality, accessibility, security, confidentiality, ownership and is stored correctly and securely. • Responsible for the provision that improvement opportunities are identified for the asset management information system. • Responsible for the control and oversight of standards, policies, processes, roles and responsibilities direct overall management of site asset information, ensuring legal and statutory obligations are met and the Glencore Copper Management of Change process is followed. • Responsible for the provision that an Asset Information Strategy is developed, implemented and updated periodically. • Responsible for the provision that asset information supports strategical, tactical and operational business processes, asset data is captured and maintained, and an asset dictionary is established and maintained. • Responsible for the provision that asset data is assessed and improved. • Provides input regarding how data is stored, acquired, archived, deleted and sustained. • Responsible for the provision that data and information is monitored and improved, to ensure data is accurate, complete, valid, consistent, unique and updated in a timely manner. • Responsible for the provision that targets are established to measure data improvement performance is acceptable and data is complaint to legal, regulatory, standards and policies. • Responsible for the provision that asset information, data, process and systems are audited, and assurance is guaranteed.
Maintenance Execution (Supervisors and Trades/Artisan)	• Responsible for the establishment and implementation of the CMMS. • Provides input into managing and assessing risk. • Responsible for the provision that information is collected, acquired and reviewed as business needs change. • Responsible for the provision that access is granted to asset information, so users are fully supported to enable them to carry out their duties, this includes the receival, use, transmission and transformation of asset information. • Responsible for the provision that asset information is maintained to ensure data quality, accessibility, security, confidentiality, ownership and is stored correctly and securely.

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	• Responsible for the provision that improvement opportunities are identified for the asset management information system. • Responsible for the provision that asset information supports strategical, tactical and operational business processes, asset data is captured and maintained, and an asset dictionary is established and maintained. • Responsible for the provision that targets are established to measure data improvement performance is acceptable and data is complaint to legal, regulatory, standards and policies. • Provides input regarding asset information, data, process and systems are audited, and assurance is guaranteed.
Supply/Warehouse	• Provides input for the establishment and implementation of the CMMS. • Provides input into managing and assessing risk. • Responsible for the provision that information is collected, acquired and reviewed as business needs change. • Responsible for the provision that access is granted to asset information, so users are fully supported to enable them to carry out their duties, this includes the receival, use, transmission and transformation of asset information. • Responsible for the provision that asset information is maintained to ensure data quality, accessibility, security, confidentiality, ownership and is stored correctly and securely. • Responsible for the provision that improvement opportunities are identified for the asset management information system. • Responsible for the provision that an Asset Information Strategy is developed, implemented and updated periodically. • Provides input regarding how data is stored, acquired, archived, deleted and sustained. • Responsible for the provision that data and information is monitored and improved, to ensure data is accurate, complete, valid, consistent, unique and updated in a timely manner. • Responsible for the provision that targets are established to measure data improvement performance is acceptable and data is complaint to legal, regulatory, standards and policies. • Responsible for the provision that asset information, data, process and systems are audited, and assurance is guaranteed.
Maintenance Planning	• Responsible for the establishment and implementation of the CMMS. • Provides input into managing and assessing risk. • Responsible for the provision that information is collected, acquired and reviewed as business needs change. • Responsible for the provision that access is granted to asset information, so users are fully supported to enable them to carry out their duties, this includes the receival, use, transmission and transformation of asset information. • Responsible for the provision that asset information is maintained to ensure data quality, accessibility, security, confidentiality, ownership and is stored correctly and securely. • Responsible for the provision that improvement opportunities are identified for the asset management information system. • Responsible for the control and oversight of standards, policies, processes, roles and responsibilities direct overall management of site asset information, ensuring legal and statutory obligations are met and the Glencore Copper Management of Change process is followed. • Provides input regarding how data is stored, acquired, archived, deleted and sustained. • Responsible for the provision that data and information is monitored and improved, to ensure data is accurate, complete, valid, consistent, unique and updated in a timely manner.

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	• Responsible for the provision that targets are established to measure data improvement performance is acceptable and data is complaint to legal, regulatory, standards and policies. • Provides input regarding asset information, data, process and systems are audited, and assurance is guaranteed.
Technical Services	• Provides input for the establishment and implementation of the CMMS. • Responsible for managing and assessing risk. • Responsible for the provision that information is collected, acquired and reviewed as business needs change. • Responsible for the provision that access is granted to asset information, so users are fully supported to enable them to carry out their duties, this includes the receival, use, transmission and transformation of asset information. • Responsible for the provision that asset information is maintained to ensure data quality, accessibility, security, confidentiality, ownership and is stored correctly and securely. • Responsible for the provision that improvement opportunities are identified for the asset management information system. • Provides input regarding how data is stored, acquired, archived, deleted and sustained. • Responsible for the provision that an Asset Information Strategy is developed, implemented and updated periodically. • Responsible for the provision that data and information is monitored and improved, to ensure data is accurate, complete, valid, consistent, unique and updated in a timely manner. • Responsible for the provision that targets are established to measure data improvement performance is acceptable and data is complaint to legal, regulatory, standards and policies. • Responsible for the provision that asset information, data, process and systems are audited, and assurance is guaranteed.
Asset Information Management	• Provides input for the establishment and implementation of the CMMS. • Provides input for managing and assessing risk. • Responsible for the provision that information is collected, acquired and reviewed as business needs change. • Responsible for the provision that access is granted to asset information, so users are fully supported to enable them to carry out their duties, this includes the receival, use, transmission and transformation of asset information. • Responsible for the provision that asset information is maintained to ensure data quality, accessibility, security, confidentiality, ownership and is stored correctly and securely. • Responsible for the provision that improvement opportunities are identified for the asset management information system. • Responsible for the control and oversight of standards, policies, processes, roles and responsibilities that direct overall management of site asset information, ensuring legal and statutory obligations are met and the Glencore Copper Management of Change process is followed. • Responsible for the provision that an Asset Information Strategy is, implemented and updated periodically and provides input into the development of the Asset Information Strategy. • Responsible for assessing, improving, storing, utilising, acquiring, archiving, deleting and sustaining data. • Responsible for the provision that targets are established to measure data improvement performance is acceptable and data is complaint to legal, regulatory, standards and policies. • Provides input with respect to the provision that asset information, data, process and systems are audited, and assurance is guaranteed.

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5. Deployment

The deployment of the Asset Information Management approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching, and mentoring

5.1 Awareness Presentation

This Guideline is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Asset Management Framework and business success.

A vital element of asset information management centres around SAP Master Data, which is the vehicle that provides the ability to hold and manage the majority of asset information within Glencore Copper. Section 5.1.1 provides a high-level overview of the main plant maintenance master data objects and their relationship in SAP.

5.1.1 Master Data Objects

Understanding master data and their relationship is fundamental to enable full functionality and capability from SAP. This understanding will amongst things, enable effective and efficient packaging of maintenance tactics, maintenance planning and reduce planning workload by enabling the system to do the heavy lifting.

There are three (3) key plant maintenance master data objects:

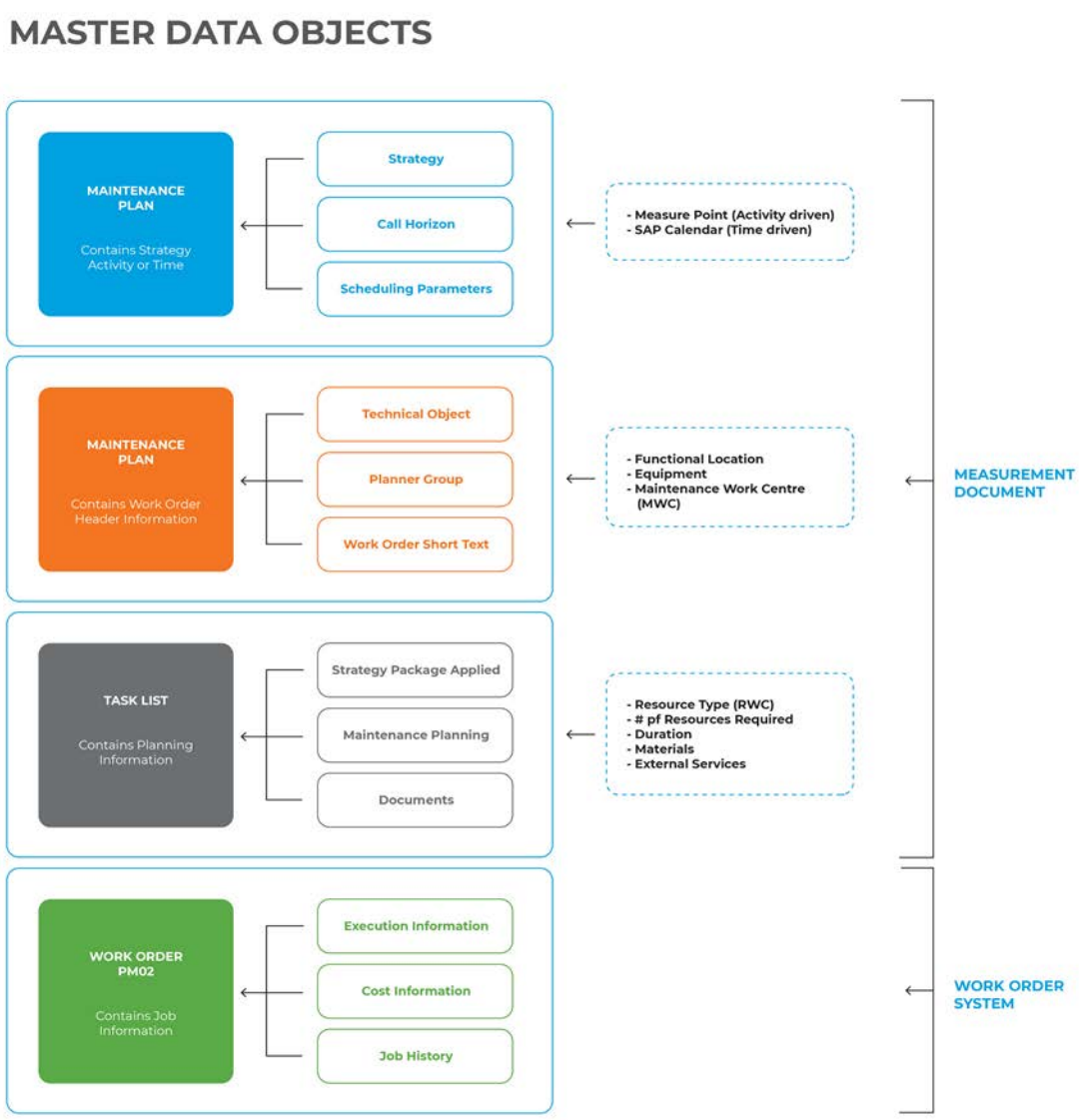
- Maintenance Plan
- Maintenance Item
- Task List

The master data objects combine to develop and deliver work orders to the work order system for planning and scheduling through the Maintenance Work Management processes as defined in Maintenance Work Management Manual (E08) and Guidelines.

Figure 5-1 provides a high-level view of the master data objects and their relationships.

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FIGURE 5-1 MASTER DATA OBJECTS



5.1.1.1 TASK LIST

Task lists are used as work templates and allow repeatable Maintenance Work Management tasks to be defined and then reused.

The key features of task lists are to identify:

- Internal labour requirements
- Labour hours
- Task duration
- External labour/service requirements
- Outline agreement (if exists)
- Component identification
- Instructions and support documents

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5.1.1.2 MAINTENANCE ITEM

Is a master data object (technical object) linked to the maintenance (work) plan which is used to call work orders (PM02), based on scheduling parameters defined in the maintenance plan.

Provides the assignment of a preventive task list to a functional location or equipment

A maintenance item includes:

- Description of task required at regular intervals
- Assignment of an SAP maintenance strategy task list and technical object to a work plan
- Assignment of multiple technical objects via an object list (currently not used in Glencore Copper)
- Information for the work order header, e.g., sort field, short text, system condition, planner group, main work centre, maintenance planning plant, planning plant and functional location

5.1.1.3 MAINTENANCE PLAN

The description of the maintenance and inspection tasks to be performed at maintenance objects. The maintenance plan describes the dates and scope of the tasks.

A maintenance plan:

- Combines information from the task list and the maintenance item
- Provides a set of packages (frequencies – calendar or activity) that define the cycle in which tasks must be performed, e.g., every 2 months or every 5000km
- Triggers regular creation of work orders (PM02) based on assigned strategies via a deadline monitoring batch job
- Provides a start date for the cycle(s) contained in maintenance item(s) and task list(s) assigned
- May be used to schedule work for one or more maintenance items
- Work order is created from each maintenance item assigned to work plan

5.2 Position Description Alignment

To effectively deploy this Guideline, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix assists in the visualisation that everyone is working towards the same goal. It clarifies, not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

5.3 Competency Assessment

This Guideline is supported by an Asset Management competencies assessment tool which is used to:

- Guide recruitment, selection, development, and deployment of people in asset management roles across the company
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance
- Map, build and manage the asset management talent pool across the company
- Identify strengths and development needs in asset management practitioners

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5.4 Training, Coaching and Mentoring

Coaching and mentoring is another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring requires addressing that a person is not currently up to the required level to perform their role as effectively and efficiently as they could. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

6. Improvement

The process to review and continually improve the approach and deployment for Asset Information Management is scheduled Annually, or when there is a sudden change in the Operating Context. As the organization matures a longer review period could be considered.

7. References

The following documents provide detailed support material that should be read in conjunction with this Standard to provide a comprehensive understanding of Asset Information Management.

TABLE 8-1 REFERENCES

DOCUMENT TITLE	DESCRIPTION
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
Asset Management Metrics Standard	This document defines the measures/KPIs, the method of calculation, the intent, and behaviours each measure is aimed at.
Institute of Asset Management Asset Information Subject Specific Guide	Defines in detail Asset Management Asset Information requirements for Asset intensive organisation

8. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Manual to support the Asset Management Framework	15/09/2021
0.2	Gather Documents and Create Draft Manual	06/10/2021
0.3	Review and Signoff 1st Draft Manual with Sites	02/11/2021
0.4	Reformat and Issue for Final Review	07/12/2021
1.0	Update Site Signoff and Publish	30/05/2022

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9. Appendix A – Stakeholder Signoff

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Anthony Green	Asset Management - Australia	Anthony.green@glencore.com.au	21/6/2022
Akeem Araoye	Asset Management – North America	Akeem.Araoye@glencore.ca	21/6/2022
Yuri Yurac	Asset Management – South America	Yuri.yurac@glencore.cl	21/6/2022
Lawrence Hlatshwayo	Asset Management – Africa	Lawrence.hlatshwayo@glencore.co.za	22/6/2022
Rohan Reisener	Asset Management - Australia	Rohan.reisener@glencore.com.au	21/6/2022
Dino Moll	Asset Management - Australia	Dino.Moll@glencore.com.au	22/6/2022

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